

OralAbsPredict

A Data-Driven Framework to Predict Human Intestinal Absorption (HIA) and Human Oral Bioavailability (HOB) from Chemical Structures

Project Level: College Project School Specification

Focus: AI/ML Drug Discovery & Cheminformatics

Core Stack: Python, RDKit, Mordred, CatBoost, SHAP, Streamlit

Status: Fully Trained, Tested (23/23 Tests Passed) & Live

1. Executive Summary: What is OralAbsPredict in Simple Words?

When scientists design new medicines, pills taken by mouth (orally) are the easiest, most convenient, and preferred route for patients. However, before an swallowed drug can treat an illness, it must accomplish two critical biological journeys:

- 1. Human Intestinal Absorption (HIA):** The pill dissolves in the stomach and must successfully pass through the gut wall into the blood vessels.
- 2. Human Oral Bioavailability (HOB):** After leaving the gut, blood flows straight into the liver. The liver tries to destroy foreign chemicals. Bioavailability is the actual percentage of active medicine that survives the liver and enters the general bloodstream.

Historically, testing this required synthesizing actual chemical powders in a lab, testing them on human gut cells (Caco-2 assays) or animals, taking weeks of labor and thousands of dollars per candidate. **OralAbsPredict** solves this problem by using **Artificial Intelligence**: scientists simply enter the chemical formula (a SMILES string), and in **less than 1 second**, the AI calculates 706 molecular parameters, predicts both HIA and HOB, and explains *why* using Explainable AI.

2. Official Problem Statement

"Poor oral absorption is one of the primary reasons for drug failure during the early stages of drug discovery, leading to increased development time and cost. Experimental evaluation of Human Intestinal Absorption (HIA) and Human Oral Bioavailability (HOB) is expensive, time-consuming, and resource-intensive. Existing computational methods often suffer from limited datasets, class imbalance, and poor interpretability. This project aims to develop an AI-driven framework that predicts HIA and HOB directly from molecular structures represented as SMILES strings. The system employs machine learning and molecular descriptors to accurately classify orally active compounds before synthesis. It also provides feature interpretation using explainable AI techniques and identifies important molecular substructures responsible for oral absorption."

3. High-Level System Architecture & Workflow

The application is built in a 5-step modular pipeline. Here is the step-by-step path a chemical molecule takes through the system:

Step 1: Input	User supplies a SMILES string (e.g. <chem>CC(=O)Oc1ccccc1C(=O)O</chem> for Aspirin) via the web interface or selects a demo drug.
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Step 2: Sanitization	RDKit parses the text, checks chemical valence rules, neutralizes formal charges, and ensures the structure is valid.
Step 3: Feature Extraction	The system calculates 706 scientific parameters (18 physical properties + 176 graph indices + 512 Morgan fingerprint bits).
Step 4: Dual AI Prediction	Two trained CatBoost machine learning models simultaneously predict: (1) Intestinal Absorption and (2) Oral Bioavailability.
Step 5: XAI & Visualization	SHAP computes exact feature contributions; RDKit draws a 2D SVG structure and computes an interactive 3D conformer.

4. Benchmark Datasets Used for Training

To ensure academic rigor, models were trained on gold-standard, peer-reviewed clinical datasets without artificial data synthesis:

Target Property	Dataset Source	Total Compounds	Positive Class (High)	Negative Class (Low)	Classification Threshold
HIA (Gut Absorption)	Hou et al. (TDC Benchmark)	578 molecules	500 (86.5%)	78 (13.5%)	Absorption >= 30%
HOB (Bioavailability)	Ma et al. Compendium	640 molecules	492 (76.9%)	148 (23.1%)	Bioavailability F >= 30%

5. Molecular Feature Engineering: What Does the AI Look At?

A computer cannot directly understand an image of a chemical. It needs numbers representing the chemical's size, shape, and chemistry. OralAbsPredict extracts **706 distinct features** for every molecule:

- 1. RDKit Physicochemical Properties (18 features):** Molecular Weight (size), LogP (oil vs water solubility / fat solubility), Topological Polar Surface Area (TPSA — area of polar atoms like oxygen/nitrogen that resist entering cell membranes), Hydrogen Bond Donors & Acceptors (HBD/HBA), Rotatable Bonds (molecule flexibility), and Lipinski's famous Rule of 5 violations.
- 2. Mordred 2D Topological Descriptors (176 features):** Graph theory parameters that capture molecular shape, branching, rings, atom connectivity, and bond polarizability without needing time-consuming 3D quantum simulations.
- 3. Morgan Circular Fingerprints (512 features):** Extended Connectivity Fingerprints (ECFP4, radius 2). Imagine breaking the molecule into circular neighborhoods of 2 bonds and checking if specific patterns (like benzene rings, alcohol groups, or ester linkages) are present (1) or absent (0).

6. Machine Learning Model Comparison: Evaluating 6 Algorithms

Rather than guessing, we trained and systematically compared all 6 major candidate algorithms on the exact same stratified holdout test sets:

Endpoint	Model Architecture	Accuracy	Balanced Acc	ROC-AUC	F1-Score	Outcome / Verdict
HIA (Gut)	CatBoost	93.97%	91.25%	0.9844	0.9645	Selected Best Model
HIA	Random Forest	93.10%	85.50%	0.9600	0.9600	Strong, slight class bias
HIA	Balanced Random Forest	90.80%	84.17%	0.9422	0.9459	Good minority sensitivity
HIA	Logistic Regression	91.95%	77.83%	0.9511	0.9542	High AUC, lower recall
HIA	LightGBM	89.66%	76.50%	0.9411	0.9404	Fast, slight overfitting
HIA	SVM (Support Vector)	90.80%	70.17%	0.9389	0.9487	Conservative classifier
HOB (Systemic)	CatBoost	74.22%	57.72%	0.7425	0.8406	Selected Best Model
HOB	Logistic Regression	70.31%	65.79%	0.7285	0.8085	Decent linear baseline
HOB	Random Forest	72.66%	63.02%	0.7039	0.8400	Competitive ensemble
HOB	LightGBM	73.44%	62.10%	0.6695	0.8497	Moderate generalization

7. Explainable AI (SHAP): Explaining the 'Why'

A major flaw of traditional machine learning in pharmaceutical science is the '**Black Box problem**'—a model says 'High' or 'Low', but chemists have no idea which atoms or properties caused the decision. OralAbsPredict uses **SHAP (SHapley Additive exPlanations)**, derived from Nobel Prize-winning cooperative game theory.

- **Local Attribution:** Every individual molecule gets an exact contribution breakdown. For example, in Aspirin, the small TPSA (63.6 Å²) and neutral formal charge strongly push the prediction toward 'High', while ester linkage slightly drags it down.
- **Key Learned Rules:**
 1. **TPSA ≤ 140 Å²:** Greatly accelerates passive cell membrane diffusion.
 2. **Neutral Species:** Fixed electric charges cause a heavy water-hydration penalty, impeding gut wall penetration.
 3. **Balanced LogP (1 to 4):** High enough to enter the fatty lipid bilayer, but low enough to avoid being trapped in fat tissue or cleared immediately by liver enzymes.

8. 2D and 3D Molecular Visualization

When an investigator enters a SMILES string, OralAbsPredict generates two rich visual representations:

1. **2D Vector Depiction:** Rendered directly via RDKit's C++ rendering engine into SVG XML. Crisp on retina screens, displaying IUPAC atom colorings, double bonds, and chiral stereocenters.
2. **3D Conformer Simulation:** RDKit uses the ETKDG (Experimental-Torsion Knowledge Distance Geometry) algorithm to create 3D atomic coordinates, followed by MMFF94 force-field energy minimization. The 3D molecule is embedded into the web app using **3Dmol.js**, allowing users to rotate, pan, and inspect the molecular shape interactively with their mouse.

9. Real-World Case Studies on Known Drugs

Compound	Drug Class	MW (g/mol)	LogP	Predicted HIA	Predicted HOB	Clinical Correlation
Paracetamol	Analgesic	151.2	1.35	High (99.4%)	High (87.6%)	Matches human oral uptake (~88%)
Aspirin	NSAID	180.2	1.31	High (99.6%)	High (80.2%)	Rapid stomach and gut absorption
Caffeine	Stimulant	194.2	-1.03	High (99.5%)	High (93.1%)	Virtually complete oral uptake (~99%)
Atorvastatin	Lipid Statin	558.7	6.31	High (82.0%)	Low (14.4%)	Correctly predicts poor bioavailability due to first-pass metabolism (~14%)

10. Academic Research & Ethical Disclaimer

Non-Clinical Research Disclaimer: OralAbsPredict is developed exclusively for computational screening, cheminformatics education, and academic research. Predictions generated by this machine learning framework do **NOT** guarantee pharmacokinetic behavior in vivo and must never be utilized as clinical diagnoses, prescribing advice, or therapeutic treatment plans. Computational predictions cannot substitute for empirical in vitro permeability assays (e.g., Caco-2, PAMPA) or in vivo human clinical trials.